Conjecture 00000004271: the rational divisible group obstructs approximation

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23 October 2026

1 The approximation obstruction

Take the actual additive group $A = (\mathbf{Q}, +)$. It is divisible: for every positive integer n and $x \in \mathbf{Q}$, the element x/n satisfies $n(x/n) = x$. The map $D(x) = 2x$ is a nonidentity additive automorphism, with inverse $x \mapsto x/2$.

On this group the identity is the only automorphism satisfying any of the following three finite-support conditions:

1. only finitely many elements are moved;
2. the image $(f - \mathrm{id})(A)$ is finite;
3. f acts identically on a subgroup of finite index.

The first is the literal finite support of a permutation. The third is the standard group-theoretic finitary condition; its equivalence with the second is stated in Dardano and Rinauro, *The group of inertial automorphisms of an abelian group*, arXiv:1403.4193v4, Introduction, pp.1–2. We use that exact preprint title and version. Lean proves all these claims for $\mathbf{Q}$, including the finite-index bridge, instead of importing the quoted equivalence as an assumption.

Thus no choice of a genuine separating metric can allow these automorphisms to approximate D. The source does not specify a formula for its "subgroup-index metric." We avoid choosing a new formula: the formal obstruction holds for *every* true metric, both for uniform approximation on A and for approximation in its automorphism group. This disproves the approximation necessary clause, so the additional proposed optimal exponent cannot rescue the stated conjunction.

2 Actual finite-support and finite-index calculations

Every additive rational automorphism satisfies

$$f(x) = f(1)x.$$

Indeed additive homomorphisms between rational vector spaces commute with rational scalar multiplication. If $f(1) \neq 1$, then

$$(f - \mathrm{id})(x) = (f(1) - 1)x$$

is injective on the infinite set $\mathbf{Q}$; its image is therefore infinite. Hence condition 2 forces $f(1) = 1$ and $f = \mathrm{id}$. Condition 1 implies condition 2 because the defect image is contained in zero together with the image of the finite moved-element set.

For condition 3 define the genuine additive subgroup

$$F = \ker(f - \mathrm{id}) = \{x \in \mathbf{Q} : f(x) = x\}.$$

If f fixes some finite-index subgroup H pointwise, then $H \leq F$, so F has finite index as well. Conversely F itself is such an H. This equivalence is proved using Mathlib's actual `AddSubgroup.FiniteIndex` and subgroup inclusion.

Let $N = [\mathbf{Q} : F]$, a positive natural number when F has finite index. The standard quotient-group calculation gives $Nx \in F$ for every $x \in \mathbf{Q}$. Therefore

$$Nf(x) = f(Nx) = Nx.$$

The nonzero rational scalar N cancels, giving $f(x) = x$ for every x. Lean checks this with the actual finite-index subgroup and `nsmul_index_mem`. It does not assume that a named "finitary" predicate has already been proved trivial. The identity satisfies all three conditions, so their union is exactly the singleton $\{\mathrm{id}\}$. In particular the equivalence of conditions 2 and 3 on this group is also a formal theorem.

3 Any genuine metric fails to provide the claimed approximation

First let d be any metric on $\mathbf{Q}$, with no compatibility assumption about its additive or field operations. Since $1 \neq 2$,

$$\varepsilon_0 = d(1, 2) > 0.$$

Every candidate f in the union above equals the identity. At $x = 1$,

$$d(f(1), D(1)) = d(1, 2) = \varepsilon_0.$$

Consequently it is impossible to find a candidate with $d(f(x), D(x)) < \varepsilon_0$ for all x. The closed Lean theorems negate the full universal approximation claim separately for literal finite support and for the standard finite-index subgroup condition.

If the source instead places the metric directly on $\mathrm{Aut}(\mathbf{Q}, +)$, let $\varepsilon_1 = d(\mathrm{id}, D) > 0$. Every candidate has distance exactly ε_1 from D, again preventing arbitrary approximation. This second interpretation is separately formalized. No formula for an undefined distance is inserted into either proof. More generally, for every T_1 topology on the actual automorphism group, the candidate set is the closed singleton $\{\mathrm{id}\}$, so D is not in its closure. This separately proved topological obstruction also covers genuine extended metrics that may take the value infinity.

This argument does not treat a nonseparating pseudometric as a metric. Nor does it identify finite rank relative to a chosen vector-space basis with finite support of group automorphisms. That different linear-algebra notion is not specified by the source and is not claimed to be refuted here.

4 Verification

Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d` are pinned. The package includes a new root build, twenty-two transitive axiom audits and full official `leanchecker --fresh Main` replay with source-bound commands, exit codes, logs and hashes. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the source, report and PDF. This is AI review, not human certification. Organizer acceptance is pending.